SESSION 12 – Matrix Multiplication & Linear Regression

Theory Answers (Tasks 4 & 5)

Task 4: Meaning of Slope and Intercept (Flipkart Price Prediction)

Context: Predicting Flipkart product price based on product rating using the linear regression equation **y = mx + c**, where y is the predicted price and x is the product rating.

Slope (m)

The slope represents how much the predicted price changes when the product rating increases by 1 unit. For example, if the slope is 200, it means that for every 1-star increase in rating, the expected price of the product rises by Rs. 200. A positive slope shows that higher-rated products tend to be priced higher, while a negative slope would mean higher ratings are associated with lower prices.

Intercept (c)

The intercept is the predicted price of a product when the rating is zero. It acts as the baseline or starting value of the price before considering the effect of rating. In real life, a zero rating may not make practical sense, but mathematically the intercept gives the model a fixed starting point so that the line can correctly fit the data.

Task 5: How Matrix Multiplication Helps in Training ML Models (Spotify)

Matrix multiplication is a core mathematical operation used while training recommendation systems like Spotify's song recommendation engine. User preferences and song features are stored in large matrices (for example, a user–song interaction matrix and a song–feature matrix). When these matrices are multiplied, the model can quickly calculate similarity scores or predicted ratings for millions of users and songs at once.

During training, matrix multiplication helps the model update its internal weights efficiently through techniques such as gradient descent. Instead of calculating one value at a time, the computer multiplies whole blocks of numbers together, making the learning process much faster. This is why modern recommendation systems can learn patterns from huge amounts of listening data and suggest songs that match a user's taste.

In short, matrix multiplication allows Spotify's model to combine user behaviour and song characteristics in a scalable way, so it can deliver personalized recommendations quickly and accurately.